

Lecture 13

- Review + Some of Exam Questions

Vector Calculus

	In	Out
$\text{grad } f$ ∇f	Scalar	vector

$\text{div } \vec{A}$ $\nabla \cdot \vec{A}$	vector	Scalar
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$\text{Curl } \vec{A}$ $\nabla \times \vec{A}$	vector	vector
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is conservative

Can

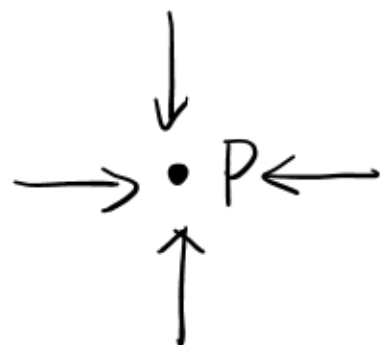
$\nabla \times (\nabla f) = 0$

if $\nabla \cdot \vec{A} = 0$ everywhere, solenoidal

if $\nabla \times \vec{A} = 0$ everywhere, conservative

$\nabla \cdot (\nabla \times \vec{A}) = 0$

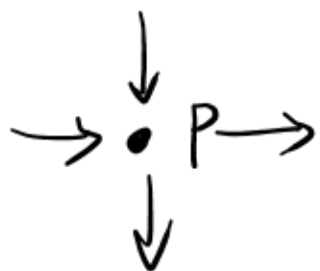
is solenoidal.



$$\operatorname{div} \vec{A} < 0$$

"sink"

$$\operatorname{curl} \vec{A} = 0$$



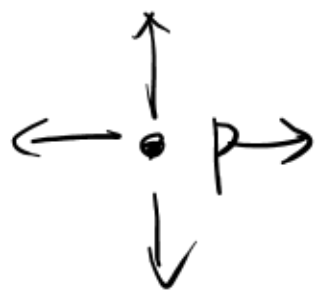
$$\operatorname{div} \vec{A} = 0$$

$$\operatorname{curl} \vec{A} = 0$$



$$\operatorname{div} \vec{A} = 0$$

$$\operatorname{curl} \vec{A} \neq 0 \quad \odot$$



$$\operatorname{div} \vec{A} > 0$$

$$\operatorname{curl} \vec{A} = 0$$

Calculate curl and divergence of the following vector field. State whether the field is conservative, solenoidal, both, or neither.

$$\vec{A} = \hat{r} 4 \exp(-5r) / r$$

Curl cares about how a vector component changes in the other directions. If A_r only changes with r , there is no variation in the θ or ϕ directions to create rotation, so $\nabla \times \mathbf{A} = 0$.

$$\boxed{\nabla \times \vec{A} = 0} \quad \text{because only } \hat{r} \text{ component \& depends on } r \text{ only}$$

$$\nabla \cdot \vec{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) = \frac{1}{r} \frac{\partial}{\partial r} (4e^{-5r}) = \frac{1}{r} (-20e^{-5r})$$

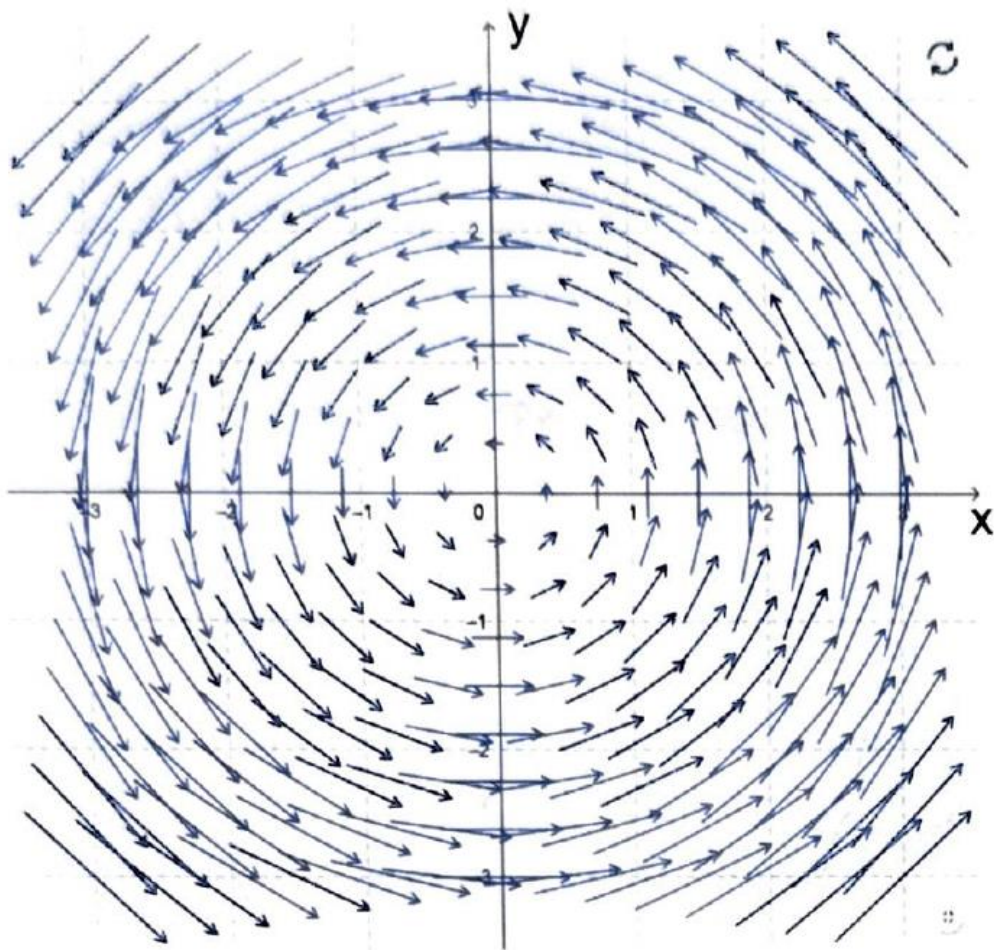
$$\Rightarrow \boxed{\nabla \cdot \vec{A} = -\frac{20}{r} e^{-5r}} \neq 0$$

+2

Field is conservative

Vector Calculus [Example 2]

a) Consider the vector field $\vec{A} = -\hat{x} y + \hat{y} x$, illustrated below. Circle the true statement(s).



(1) The field is conservative

$$\nabla \cdot \vec{A} = 0$$

☒ (2) The field is solenoidal

$$\nabla \times \vec{A} =$$

(3) $\nabla \times \vec{A}$ at the origin is zero

$$\hat{z} \left(\frac{\partial}{\partial x} \times -\frac{\partial}{\partial y} (-y) \right)$$

☒ (4) $\nabla \times \vec{A}$ at the origin points in $\hat{z}$

$$= 2\hat{z}$$

(5) $\nabla \times \vec{A}$ at the origin points in $-\hat{z}$

☐ (6) $\nabla \cdot \vec{A}$ at the origin is zero

(7) $\nabla \cdot \vec{A}$ at the origin is positive

(8) $\nabla \cdot \vec{A}$ at the origin is negative

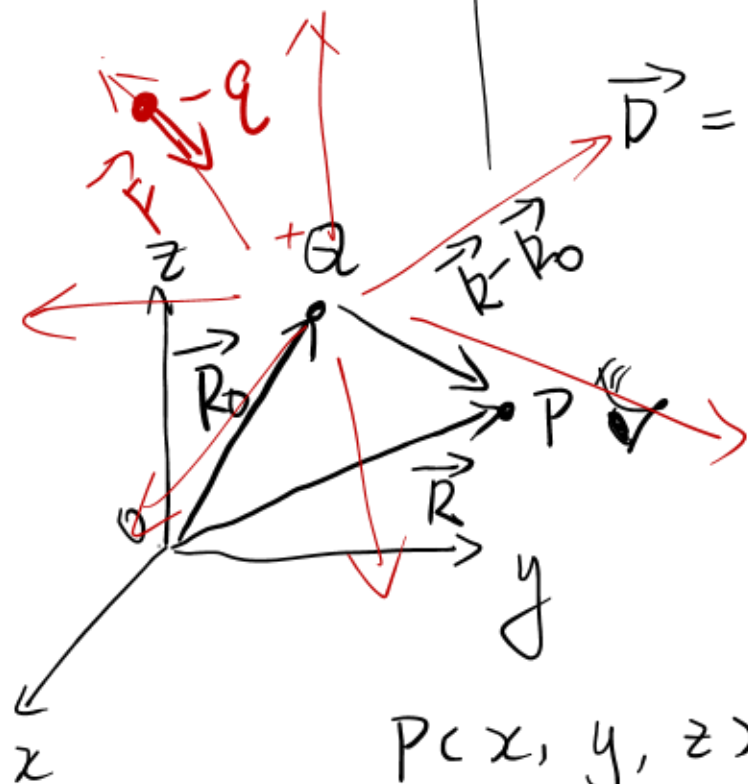
Maxwell's Eqns for electrostatic field

$$\nabla \cdot \vec{D} = \rho_v$$

$$\nabla \times \vec{E} = 0$$

 $\Rightarrow \vec{E}$ is conservative $\Rightarrow V$

$$\vec{D} = \epsilon \vec{E}$$



$$\vec{E} = \frac{Q}{4\pi\epsilon |\vec{R} - \vec{R}_0|^3} (\vec{R} - \vec{R}_0)$$

$$\vec{R} = \vec{R}_0 + (\vec{R} - \vec{R}_0)$$

$$P(x, y, z)$$

$$P(r, \phi, z)$$

$$P(R, \theta, \phi)$$

$$\vec{R} = \hat{x}x + \hat{y}y + \hat{z}z$$

$$\vec{R} = \hat{r}r + \hat{z}z$$

$$\vec{R} = \hat{R}R$$

$$\vec{F} = q\vec{E}$$

Electric field

How to cal. $\vec{E}$ from charge distributions?

general: Coulomb's law + superposition

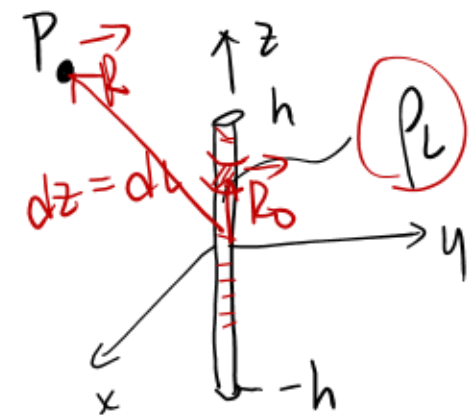


symmetric charge distribution

Gauss's law.

Partition the source into many point-like charges $dq(\vec{R}_0)$

cal $d\vec{E}$, then apply superposition to sum/integrate $d\vec{E}$ from all the charges.



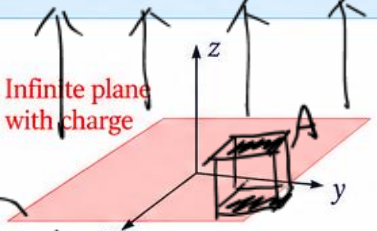
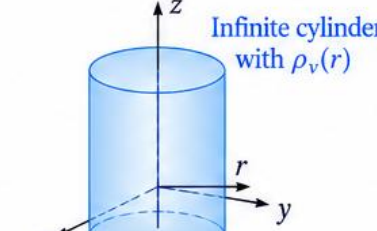
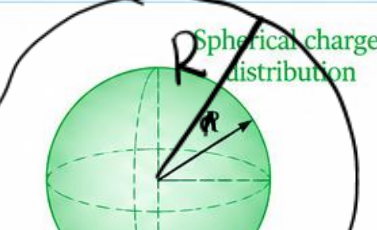
$$dq = \rho_L dz \Rightarrow d\vec{E} = \frac{\rho_L dz}{4\pi\epsilon |\vec{R} - \vec{R}_0|^3} (\vec{R} - \vec{R}_0)$$

$$\Downarrow$$
$$\vec{E} = \int_{z=-h}^h d\vec{E}$$

* write down
 dz , ds , dv
in diff. systems.

Electric field (symmetric charge distribution) -- Gauss's law

$$\oint \vec{D} \cdot d\vec{S} = Q_{enc}$$

Charge distribution	Geometry / illustration	Symmetry test	Electric field form
1. Infinite plane charge	 Infinite plane with charge ρ_s	Translate anywhere parallel to the plane → unchanged	$E = E(z)\hat{z}$ (normal to plane) $D \cdot A + D \cdot A = \rho_s \cdot A$ $D = \rho_s / 2 \Rightarrow \vec{E} = \frac{\vec{D}}{\epsilon} = \pm \hat{z} \frac{\rho_s}{2\epsilon}$
2. Infinite line charge	 Infinite line with charge ρ_L	Rotate about the line and translate along the line → unchanged	$E = E(r)\hat{r}$ (radially outward) $D_r \cdot 2\pi r h = \rho_L h$ $D_r = \frac{\rho_L}{2\pi r}$
3. Infinite cylindrical charge distribution (volume charge density $\rho_v = \rho_v(r)$)	 Infinite cylinder with $\rho_v(r)$	Rotate about the axis and translate along the axis → unchanged	$E = E(r)\hat{r}$
4. Spherical charge distribution (volume charge density $\rho_v = \rho_v(R)$, including point charge)	 Spherical charge distribution	Rotate about the center → unchanged	$E = E(R)\hat{R}$

$$D \cdot \oint dS = Q_{enc}$$

Surface area

$$D_r = \frac{\rho_L}{2\pi r}$$

$$E_r = \frac{\rho_L}{2\pi \epsilon r}$$

$$\vec{E} = \hat{r} \frac{\rho_L}{2\pi \epsilon r}$$

$$D_R \cdot 4\pi R^2 = Q_{enc}$$

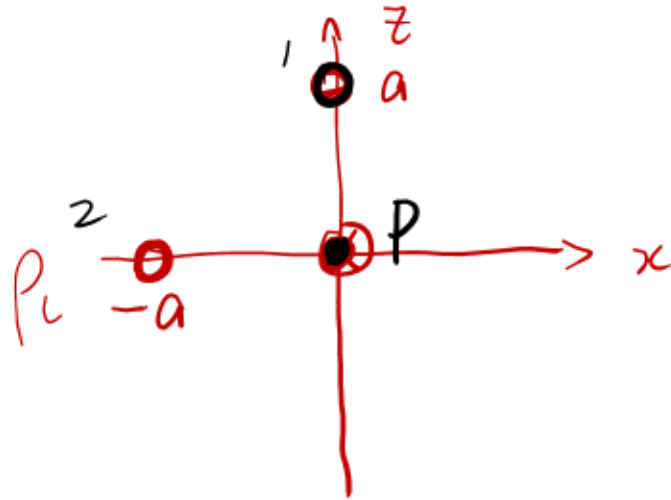
$$D_R = \frac{Q_{enc}}{4\pi R^2}$$

$$E_R = \hat{R} \frac{Q_{enc}}{4\pi \epsilon R^2}$$

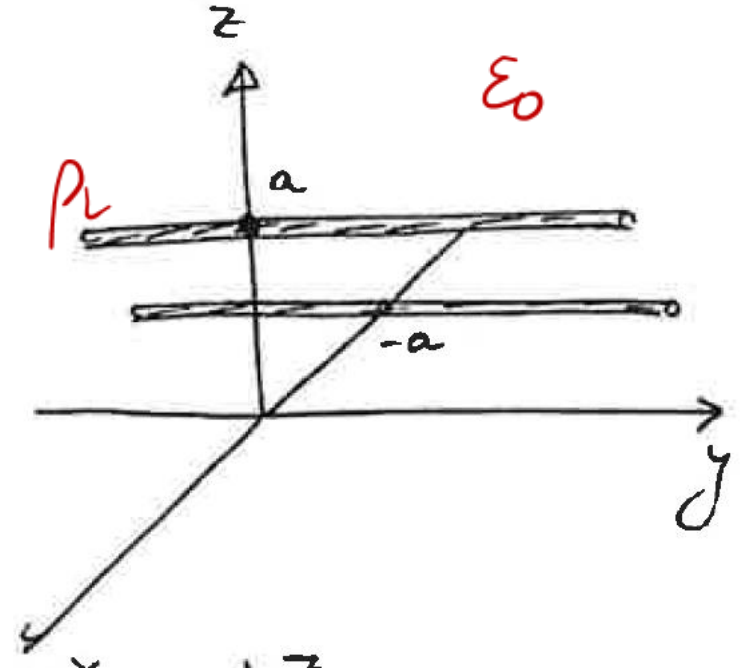
Electric field [Example]

Consider two infinitely long wires, both parallel to the y -axis, one located at $x = -a$ and $z = 0$, the other at $x = 0$ and $z = a$ (see picture). The two wires have the same line charge density ρ_L . The material throughout is vacuum.

- a) Write down the electric field vector $\vec{E}$ at any point P on the y -axis (i.e., for points that have $x = 0$ and $z = 0$).



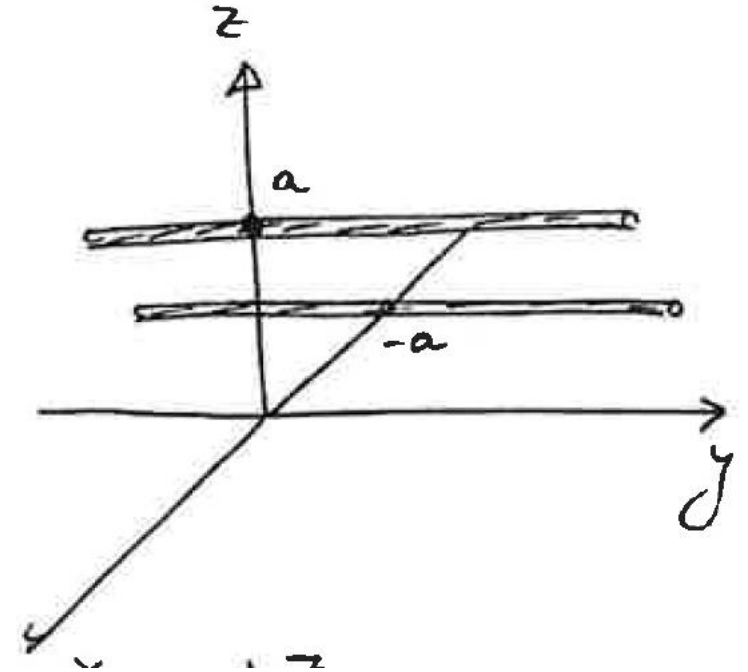
$$\vec{E} = \hat{r} \frac{\rho_L}{2\pi\epsilon r}$$
$$\vec{E}_1 = -\hat{z} \frac{\rho_L}{2\pi\epsilon a}$$
$$\vec{E}_2 = \hat{x} \frac{\rho_L}{2\pi\epsilon a}$$
$$\vec{E} = \frac{\rho_L}{2\pi\epsilon a} (\hat{x} - \hat{z})$$



Electric field [Example]

Consider two infinitely long wires, both parallel to the y -axis, one located at $x = -a$ and $z = 0$, the other at $x = 0$ and $z = a$ (see picture). The two wires have the same line charge density ρ_l . The material throughout is vacuum.

- a) Write down the electric field vector $\vec{E}$ at any point P on the y -axis (i.e., for points that have $x = 0$ and $z = 0$).



If charge Q is placed at the coordinate origin, what is the force $\vec{F}$ (magnitude and direction) it feels from the two wires? Remember that the force on charge Q is related to the electric field at that point as $\vec{F} = Q\vec{E}$.

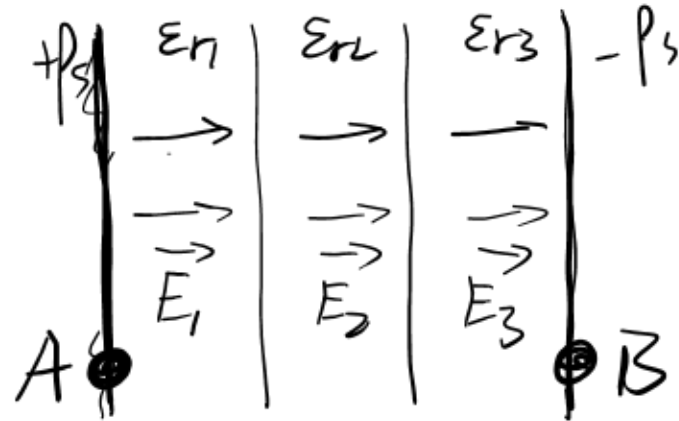
Potential - scalar

$$\nabla \times \vec{E} = 0 \quad \vec{E} \text{ is conservative} \Rightarrow V$$

Potential difference:-

$$V_A - V_B = - \int_B^A \vec{E} \cdot d\vec{l}$$

any path you'd like



Potential requires a ref. point $V_{ref} = 0$
 method 1 = if you already cal. $\vec{E}$

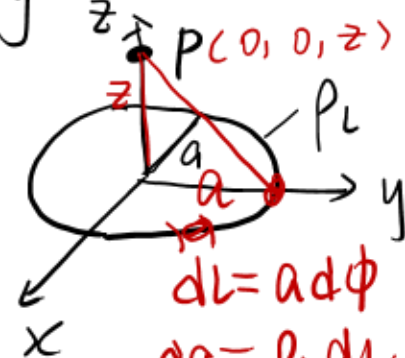
$$V_P = - \int_{ref}^P \vec{E} \cdot d\vec{l}$$

finite charge distribution $V_{\infty} = 0$

Method 2 =

$$V = \frac{Q}{4\pi\epsilon |\vec{r} - \vec{r}_0|}$$

$$dV = \frac{\rho_L a d\phi}{4\pi\epsilon \sqrt{a^2 + z^2}}$$



$$dq = \rho_L dL = \rho_L a d\phi$$

$$V = \int_{\phi=0}^{2\pi} \frac{\rho_L a d\phi}{4\pi\epsilon \sqrt{a^2 + z^2}} = \frac{\rho_L a}{2\epsilon \sqrt{a^2 + z^2}}$$

Electrical properties of materials

In the presence of external electric field

- Dielectrics polarize

$$\begin{aligned}\vec{D} &= \epsilon_0 \vec{E} + \vec{P} \\ &= \epsilon_0 (1 + \chi_e) \vec{E} \\ &= \epsilon_0 \epsilon_r \vec{E}\end{aligned}$$

$$\boxed{\vec{D} = \epsilon \vec{E}}$$

$$\vec{P} = \chi_e \epsilon_0 \vec{E}$$

$\epsilon_r = 1 + \chi_e$

$$\epsilon = \epsilon_0 \epsilon_r$$

$\times$

$\epsilon_r = 2$

$$\begin{aligned}\vec{D} &= \hat{R} D_0 \\ \vec{E} &= \frac{\vec{D}}{\epsilon_0 \epsilon_r} = \frac{\vec{D}}{2\epsilon_0}\end{aligned}$$

$\chi_e = 1$

Review: Perfect conductor (PEC) in electrostatic equilibrium $(\sigma \rightarrow \infty)$

Note:-

$$\textcircled{1} \quad \vec{J} = \sigma \vec{E} \quad \vec{E} = \frac{\vec{J}}{\sigma} \rightarrow 0 \quad \text{Inside of PEC, } \vec{E} = 0$$

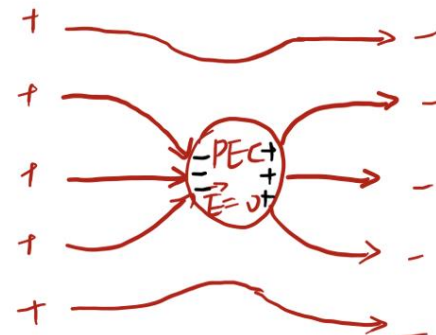
$$\vec{B} = 0 \quad \rightarrow \quad \nabla \cdot \vec{D} = \rho_v$$

② Inside of PEC, net volume charge density ρ_v is 0 everywhere.

- Where do the free charges go?
- The free charges distribute themselves to the surface, giving rise to the surface charge density ρ_s .

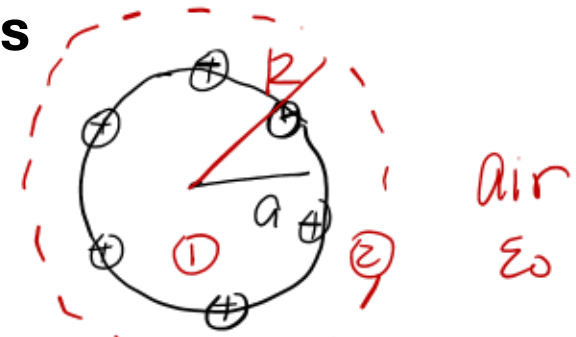
③ PEC is equipotential surface. $A \text{---} \vec{E}=0 \text{---} B \quad V_A = V_B$

④ $\vec{E}_{\text{external}}$ is always normal to PEC surface



Exam questions related to electrical properties of materials

Charge Q has been placed on a metallic sphere of radius a (i.e., the sphere is made of an excellent conductor). The surrounding medium is air (relative permittivity $\epsilon_r = 1$).



Region ① $R < a$ $\vec{E} = 0$ $\vec{D} = 0$

Region ② $R > a$

$$\oint_S \vec{D} \cdot d\vec{S} = Q_{enc}$$

$$D_R \cdot 4\pi R^2 = Q$$

$$D_R = \frac{Q}{4\pi R^2}$$

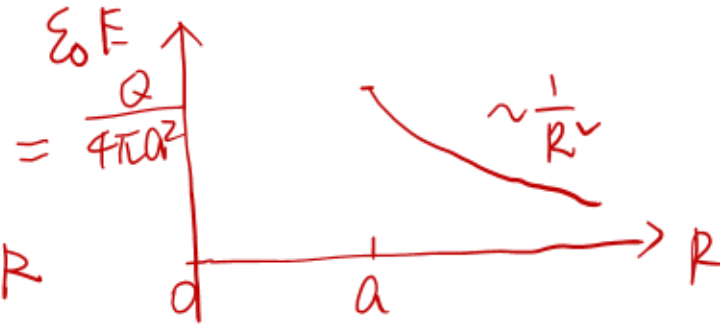
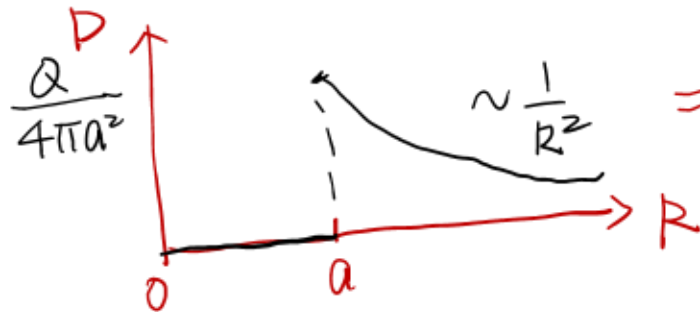
a) Use Gauss's law to calculate the electric flux density $\vec{D}$ and electric field intensity $\vec{E}$ everywhere in space. Graph $D(R)$ vs R and $\epsilon_0 E(R)$ vs R . (Hint: The sphere is metallic. That should guide your assumptions about how the charge is distributed.)

b) A thick dielectric spherical shell is added to the system (i.e., in addition to the charged sphere from before). The dielectric extends from $R = a$ to $R = b$, and has relative permittivity of $\epsilon_r = 2$. Calculate and graph $D(R)$ vs R , $\epsilon_0 E(R)$ vs R , and $P(R)$ vs R .

c) Calculate the potential difference $V(a) - V(b)$ for the setup described in part (b) (i.e., both the conductor and the dielectric shell together).

$$\vec{D} = \hat{R} \frac{Q}{4\pi R^2}$$

$$\vec{E} = \hat{R} \frac{Q}{4\pi \epsilon_0 R^2}$$

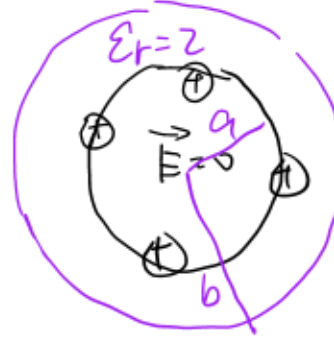


Exam questions related to electrical properties of materials

$$V(a) - V(b) = - \int_b^a \vec{E} \cdot \hat{R} dR$$

Charge Q has been placed on a metallic sphere of radius a (i.e., the sphere is made of an excellent conductor). The surrounding medium is air (relative permittivity $\epsilon_r = 1$).

a) Use Gauss's law to calculate the electric flux density $\vec{D}$ and electric field intensity $\vec{E}$ everywhere in space. Graph $D(R)$ vs R and $\epsilon_0 E(R)$ vs R . (Hint: The sphere is metallic. That should guide your assumptions about how the charge is distributed.)



$$= - \int_b^a \frac{Q}{8\pi\epsilon_0 R^2} dR$$

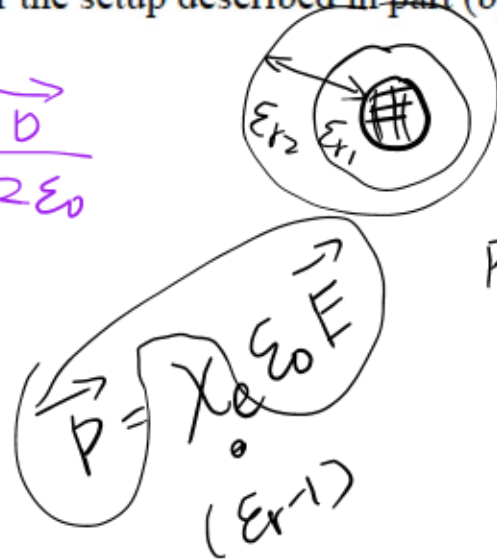
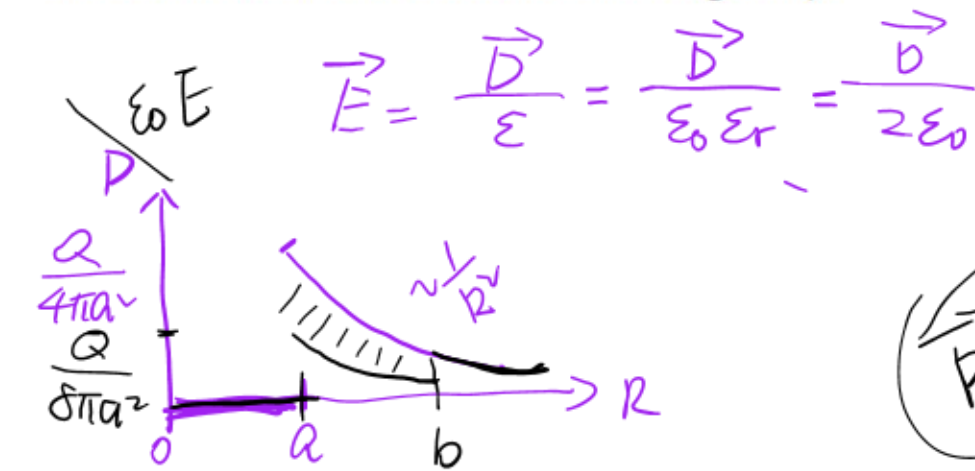
$$= - \frac{Q}{8\pi\epsilon_0} \left(-\frac{1}{R} \right) \Big|_b^a$$

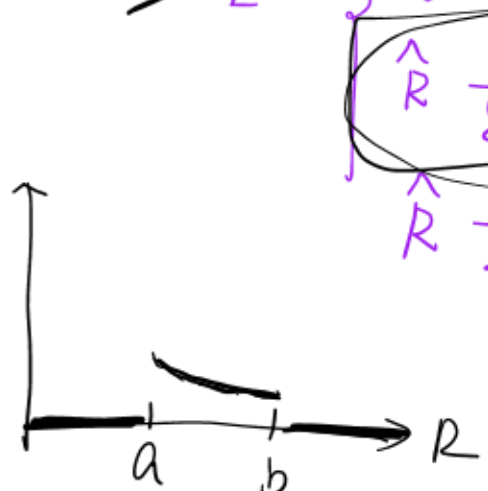
b) A thick dielectric spherical shell is added to the system (i.e., in addition to the charged sphere from before). The dielectric extends from $R = a$ to $R = b$, and has relative permittivity of $\epsilon_r = 2$. Calculate and graph $D(R)$ vs R , $\epsilon_0 E(R)$ vs R , and $P(R)$ vs R .

$$\vec{D} = \begin{cases} 0 & R < a \\ \hat{R} \frac{Q}{4\pi R^2} & R > a \end{cases}$$

$$= \frac{Q}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$$

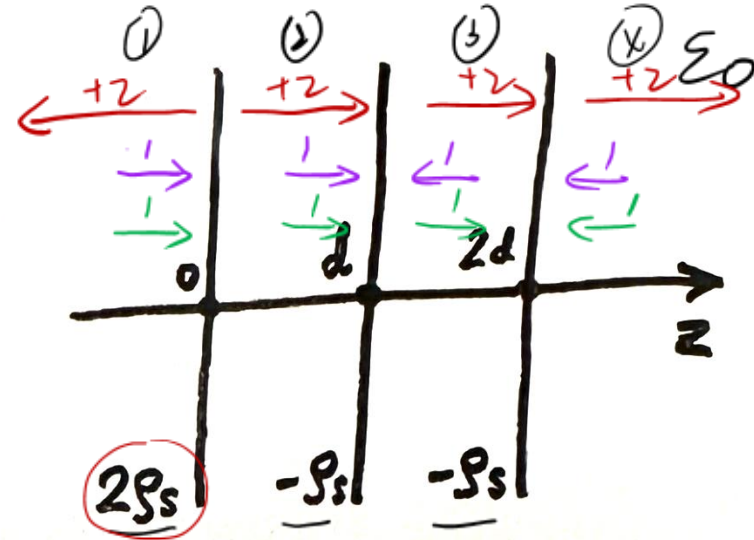
c) Calculate the potential difference $V(a) - V(b)$ for the setup described in part (b) (i.e., both the conductor and the dielectric shell together).



$$\Rightarrow \vec{E} = \begin{cases} 0 & R < a \\ \hat{R} \frac{Q}{8\pi\epsilon_0 R^2} & a < R < b \\ \hat{R} \frac{Q}{4\pi\epsilon_0 R^2} & R > b \end{cases}$$


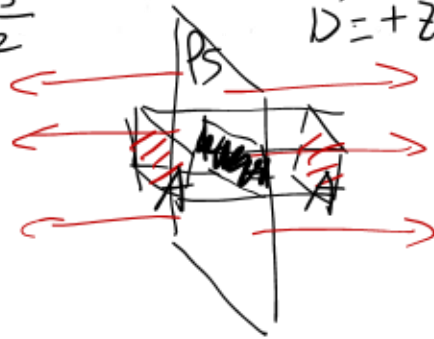
Exam questions related to electrical properties of materials

Problem 3: Three infinite parallel planes have surface charge densities $2\rho_s$, $-\rho_s$, $-\rho_s$ ($\rho_s > 0$). They are normal to the z -axis and placed at $z=0$, $z=d$, and $z=2d$, respectively, as in the figure. (Read carefully. This is NOT the same as the practice midterm!). Assume the planes are in vacuum (permittivity ϵ_0).



- a) Find $\vec{E}$ everywhere in space. Graph $E(z)$ versus z .
- b) Calculate and graph the electric potential $V(z)$. Assume $V(0) = 0$.

$$\vec{D} = -\hat{z} \frac{\rho_s}{2}$$



$$\vec{D} = +\hat{z} \frac{\rho_s}{2}$$

$$\oint_S \vec{D} \cdot d\vec{S}$$

$$2D_z \cdot A = \rho_s A$$

$$D_z = \frac{\rho_s}{2}$$

$$\rho_s \Rightarrow |\vec{D}| = \frac{\rho_s}{2}$$

Region ① $z < 0$, $\vec{E} = 0$

② $0 < z < d$, $\vec{E} = 4 \frac{\rho_s}{2\epsilon_0} \hat{z}$

$$= \hat{z} \frac{2\rho_s}{\epsilon_0}$$

③ $d < z < 2d$

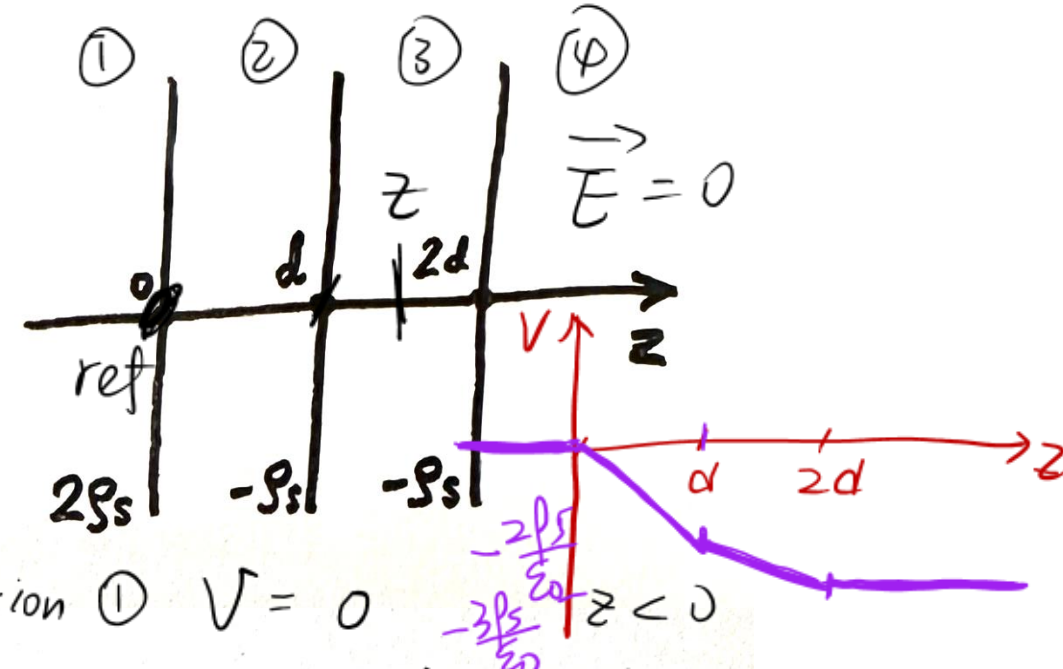
$$\vec{E} = \hat{z} \cdot 2 \cdot \frac{\rho_s}{2\epsilon_0} = \hat{z} \frac{\rho_s}{\epsilon_0}$$

④ $z > 2d$

$$\vec{E} = 0$$

Exam questions related to electrical properties of materials

Problem 3: Three infinite parallel planes have surface charge densities $2\rho_s$, $-\rho_s$, $-\rho_s$ ($\rho_s > 0$). They are normal to the z -axis and placed at $z=0$, $z=d$, and $z=2d$, respectively, as in the figure. (Read carefully. This is NOT the same as the practice midterm!). Assume the planes are in vacuum (permittivity ϵ_0).



a) Find $\vec{E}$ everywhere in space. Graph $E(z)$ versus z .

b) Calculate and graph the electric potential $V(z)$. Assume $V(0)=0$.

$$\vec{E} = \begin{cases} 0 & z < 0 \\ \vec{E}_2 = \hat{z} \frac{2\rho_s}{\epsilon_0} & 0 < z < d \\ \vec{E}_3 = \hat{z} \frac{\rho_s}{\epsilon_0} & d < z < 2d \\ 0 & d > z > 2d \end{cases}$$

Region ④ $V = -\frac{3\rho_s}{\epsilon_0} d$

Region ① $V = 0$

Region ② $V = -\int_{z=0}^z \frac{2\rho_s}{\epsilon_0} dz$

Region ③

$$V = -\int_{z=0}^d \vec{E}_2 \cdot d\vec{l} - \int_{z=d}^z \vec{E}_3 \cdot d\vec{l}$$

$$= -\frac{2\rho_s}{\epsilon_0} d - \left(\frac{\rho_s}{\epsilon_0} z - \frac{\rho_s}{\epsilon_0} d \right)$$

$$= -\frac{\rho_s}{\epsilon_0} d - \frac{\rho_s}{\epsilon_0} z$$